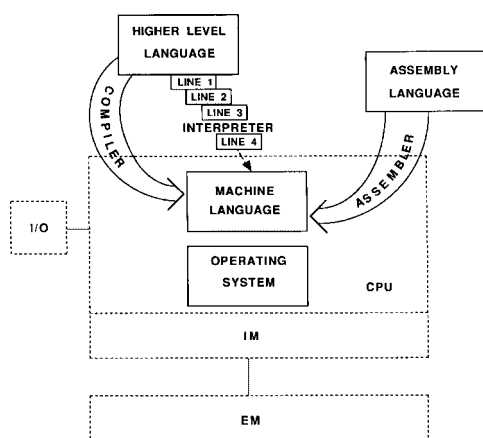
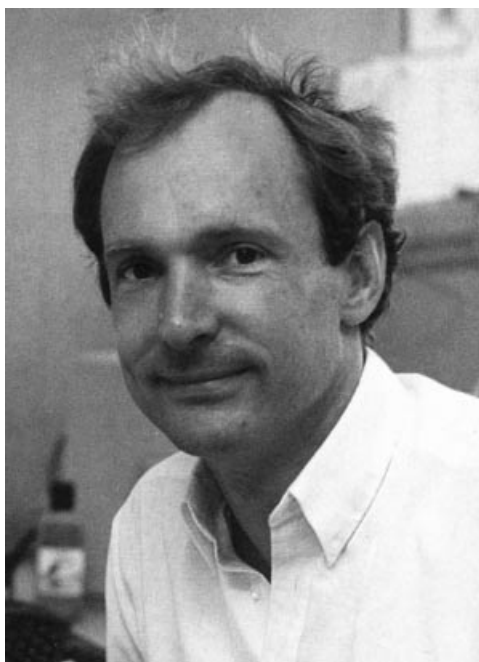


**MARGIN FIGURE 10-20**

Relationships among software and hardware in a typical computer. Hardware is shown inside dashed lines. I/O = input/output; CPU = central processing unit; IM = internal memory; EM = external memory.



History of the Internet: The World Wide Web was created in March 1989, at CERN, a high-energy particle physics laboratory on the Franco-Swiss border. Tim Berners-Lee, a physicist at CERN, proposed the idea of using a hypertext system that would link data from diverse information sources and different computer platforms.

streams of digital data, even if a number of its components or linkages were destroyed. The solution was “packet switching” a system in which data streams are broken into smaller pieces, called cells or frames (see Margin Figure 10-18). Each cell contains not only a piece of the data, but also information such as the place of the cell in the original sequence, the priority of this stream of data compared with other data streams, and so forth. One key feature of packet switching is that each cell also contains the address of the component to which it is being sent. Another key feature of packet switching is that the network consists of interconnected routers. Each router is a computer whose purpose is to maintain information about the addresses of surrounding routers. When a packet arrives at a router, it is automatically sent on to another router that is “closer” to its destination. Thus, if part of the network is disabled, the routers update their information and simply send cells via different routes. This earliest wide-area network was known as Arpanet, after the Department of Defense’s Advanced Research Projects Administration.

When electronic mail (e-mail) was developed, the usefulness of Arpanet became apparent to a wider community of users in academics and in government, and the number of network users continued to grow. Arpanet was officially decommissioned in 1989. By that time, a wide community of users required that the network be continued. The administration of the network was turned over the National Science Foundation. The network of routers, file servers, and other devices that have become the foundation of modern communications has been known as the Internet since 1989.

In 1989, the World Wide Web (WWW) was first proposed at CERN, a high-energy physics laboratory on the border between France and Switzerland. This proposal embodied all of the modern components of the WWW, including file servers that maintain “pages” of data in standard format with “hypertext” links to other sources of data or other servers. In 1991 there were only 10 file servers on the WWW, at various physics laboratories. Today, the number of servers is well into the millions.

Network Components and Structure

Computer networks for medical imaging are known by many names, such as information management, archiving, and communications systems (IMACS), picture archiving and communications systems (PACS), digital imaging networks (DIN), and local area networks (LAN). These networks face some fundamental problems, including (1) how to transmit images quickly, (2) how to avoid bottlenecks or “data collision” when the network is experiencing heavy use, (3) how to organize and maintain the data base or “log” that records the existence of images and their locations on the system, and (4) how to retain as many images as possible on the network for as long as necessary.

The main components of a computer network and its typical organization are shown in Margin Figure 10-21. Components include image acquisition (CT, ultrasound, etc.), archiving (tape, disk, etc.), central controller, data base manager, and display station. Components that allow the network to communicate with the outside world through display and archiving techniques are referred to as “nodes.” All components are not necessarily nodes. For example, several ultrasound units might be connected to a single formatting device that translates the digital images into a standard format recognizable by the network. The formatting device communicates directly with the network and is the node for the ultrasound units.

Networks can be divided into two categories on the basis of their overall structure. Centralized networks use a single computer (a central controller) to monitor and control access to information for all parts of the network. A distributed network contains components that are connected together with no central controller. Tasks are handled as requested until a conflict arises such as more demands being placed on a component than it can handle. While it may appear chaotic, a distributed network has the advantage that other components are not affected by slowdown or stoppage

of one or more components of the system. In particular, there is no central controller that would shut the whole system down if it were to malfunction.

Interfaces

The interface of an imaging device such as a CT scanner with a computer network is usually a more complex matter than simply connecting a few wires. Transfer of images and other data requires that the network must be ready to receive, transmit, and store information from the device. These operations could interfere with other activities taking place on the network. Therefore, the imaging device may have to send an “interrupt” signal that indicates that a transmission is ready. The data may have to be transmitted in blocks of specified size (e.g., 256-Kbyte transmission packets) that are recognizable to the network as part of a larger file of information.

The above-mentioned problems require both hardware and software solutions. Programs that control transmission of data to and from various components are called “device drivers.” The physical connectors must also be compatible with (i.e., can be attached and transmit signals to) other network components. Both hardware and software are implied in the term “interface.” Connecting a component to a computer or computer network in such a way that information may be transmitted is described as interfacing the component. There have been a number of attempts to standardize interfaces for imaging devices, including the American College of Radiology/National Electronics Manufacturers Association (ACR/NEMA) standards.¹⁷

Transmission Media

Components of a network may be physically separated by distances varying from a few feet to several hundred miles or more. The transmission of digital information requires a transmission medium that is appropriate to the demands of the particular network. One of the most important factors to consider is the rate at which data are transmitted, measured in bits per second (bps).

Example 10-5

A network is capable of transmitting data at a rate of 1 megabit per second (Mbps). If each pixel has a bit depth of 8 bits, how long will it take to transmit 50 512 × 512 images?

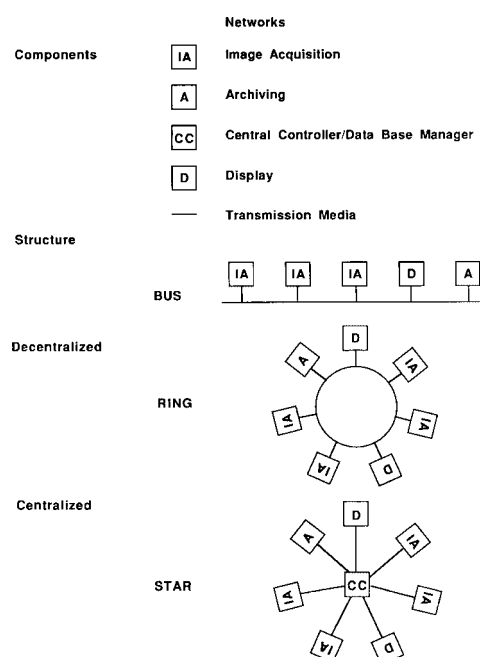
Because each pixel consists of 8 bits, the total number of bits is

$$\begin{aligned} 50[8 \text{ bits/pixel} \times (512 \times 512 \text{ pixels})] &= 50(2^3 \times 2^9 \times 2^9) \text{ bits} \\ &= 50(2^{21}) \text{ bits} \\ &= (50)(2 \text{ Mbits}) = 100 \text{ Mbits} \end{aligned}$$

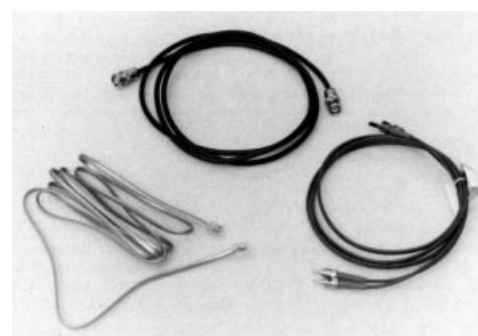
The transmission time is then

$$\frac{100 \text{ Mbits}}{1 \text{ Mbps}} = 100 \text{ sec}$$

One of the least expensive transmission media is telephone wire (sometimes called “twisted pairs”). It is inexpensive and easy to install and maintain. However, the transmission rate does not usually exceed a few Mbps. Higher rates up to hundreds of Mbps are achievable with coaxial cable. However, coaxial cable is more expensive, needs inline amplifiers, and is subject to electrical interference problems in some installations. The highest transmission rates are achievable with fiber-optic cable. This transmission medium consists of glass fibers that transmit light pulses, thereby eliminating electrical interference problems. Transmission rates for fiber-optic cables are currently just below a terabit per second (Tbps). Telephone, coaxial cable, and fiber-optic cable are shown in Margin Figure 10-22.



MARGIN FIGURE 10-21
Components and structures commonly used in image networks.



MARGIN FIGURE 10-22
Transmission media used in imaging networks. Clockwise from the top: coaxial cable, fiber-optic cable, unshielded “twisted pair” (telephone cable).

Digital Communications Options*Copper Wire:*

Modulator/Demodulator (modem)	56 kbps
-------------------------------	---------

Integrated Services Digital Network (ISDN)	64 or 128 kbps
--	----------------

Digital Subscriber

Line (DSL)	Several Versions up to 4 Mbps
Asynchronous DSL	3-4 Mbps to the home <1 Mbps from the home

Hybrid-Fiber/Coax:

Cable modem	40 MBps to the home,
	However, multiple users slow it down
	Usually, ≤ 10 MBps from cable box to your computer

*Wireless (high frequency GHz radio waves):**Local Multi-Point*

Distributed Services (LMDS)	data rates of up to 155 Mbps problems: "rain fade", trees, etc.
-----------------------------	--

Cellular Telephone	Now: 10 to 50 kbps Future: 1 Mbps
--------------------	--------------------------------------

Satellites	data rates of up to 210 Mbps problem: cost
------------	---

Direct Fiber:

fiber-to-the-home	Now: <10 Mbps Ethernet for most PCs <100 Mbps FDDI for some devices 10's of Gbps Asynchronous Transfer Mode Future: Tbps
-------------------	---

Data Compression

Images are transmitted faster and require less storage space if they are composed of fewer bits. Decreasing the number of pixels reduces spatial resolution, however, and decreasing the bit depth decreases contrast sensitivity. It is possible, however, to "compress" image data in such a way that fewer bits of information are needed without significant loss of spatial or contrast resolution.

One way to reduce the number of bits is to encode pixel values in some sequence (e.g., row by row) to indicate the value of each pixel and the number of succeeding pixels with the same value. This decreases the total number of bits needed to describe the image, because most images have several contiguous pixels with identical values (e.g., the black border surrounding a typical CT image). Other techniques for data compression include (a) analysis of the probability of occurrence of each pixel value and (b) assignment of a code to translate each pixel value to its corresponding probability. Such a "probability mapping" uses fewer bits for the more probable pixel values.¹⁹

In the examples above, the full information content of the image is preserved. When the image is expanded, it is exactly the same as it was before compression. These approaches to data compression are known as lossless (bit-preserving, or nondestructive) techniques. They may yield a reduction in the number of required bits by a factor of 3 or 4. When a greater reduction is needed, methods of data compression may be used that do not preserve the exact bit structure of an image, but still maintain acceptable diagnostic quality for the particular application. These techniques, known as irreversible (non-bit-preserving or destructive) data compression methods, can reduce the number of bits by an arbitrarily high factor.¹²

An example of an irreversible data compression method involves the use of the Fourier transform to describe blocks of pixel values in the image. Some of the high- or low-frequency components of the image are then eliminated to reduce the number of bits required to store the image. When an inverse transform is used to restore the image, the loss of high or low spatial frequencies may not significantly detract from the diagnostic usefulness of the image.

Display Stations and Standards

The part of a computer network that is most accessible to the observer is the digital display or monitor, usually referred to as the image workstation. Some display stations are capable of displaying more data than are presented on the screen at one time. They may, initially, show images at reduced resolution (e.g., 1024×1024) while preserving the full "high-resolution" data set (e.g., 2048×2048) in memory. The stored data can be recalled in sections through user-selectable windows (analogous to using a magnifying glass on a part of the image). Alternately, only part of the image may be presented, but the part that is presented may be "panned" or moved around over the full image.

Standards for digital matrix size and remote display of medical images that are acceptable for primary diagnosis from computer monitors have been published by the Radiological Society of North America.¹⁸ These standards will continue to evolve as equipment performance (particularly display monitors) continues to improve. Currently, two classes of images are recognized: small matrix systems and large matrix systems. Small matrix systems (CT, MRI, ultrasound, nuclear medicine, and digital fluoroscopy) must have a format of at least $5 \text{ k} \times 5 \text{ k} \times 8$ bits. The display must be capable of displaying at least $5 \text{ k} \times 0.48 \text{ k} \times 8$ bits. Large matrix systems (digitized radiographs, computed radiography) are held to a standard based upon required spatial and contrast resolution. For those imaging methods, the digital data must provide a resolution of at least 2.5 line pairs per millimeter and a 10-bit gray scale. The display must be capable of resolving 2.5 line pairs per millimeter with a minimum 8-bit gray scale.

The industry standard format for transferring images between components of a network is the Digital Imaging and Communications in Medicine or DICOM standard. The standard consists of specifications for various data “fields” that must occur in the image header. In addition to pixel values, certain patient data should accompany each image. For example, each image should be preceded by “header” information that gives patient demographic data as well as details of the structure of data in the image file. These fields describe attributes such as the matrix size of the image, whether it is part of a series (e.g., one slice of a multislice CT series), and patient demographic data.

Although the capabilities of “high-end” workstations are far from being standardized, some general features have been. A monitor should be able to display enough written information concerning the patient to obviate the need for transport of paper documents. Obviously, it must be able to display text and perform at least rudimentary editing and file organization functions. The display station should be able to run image processing software as well as provide simple display functions such as variable window level, window width, and magnification. Dimensional reconstruction and tissue segmentation (display of bones only or vessels only) are becoming more common.

The development of the DICOM standard was initiated by the American College of Radiology (ACR) and the National Electrical Manufacturer's Association (NEMA) in 1985, and it continues to evolve as equipment capabilities change.

PROBLEMS

- *10-1. How many bits of information are contained in 5 kilobits (kbits)?
- *10-2. How many bits of information are contained in 60 megabytes (Mbytes)? How many bytes?
- *10-3. How many bits of information are contained in 0.5 Mwords if a 32-bit word length is used? How many bytes?
- *10-4. A $512 \times 512 \times 12$ -bit image is transmitted along fiber-optic lines at a rate of 1 gigabit/sec (1 Gbit = 2^{30} bit). How many of these images can be transmitted per minute at such a rate? (Note: Demographic and other textual information would accompany each image, reducing the actual rate of transmission.)
- *10-5. A 4-bit A to D converter is used to record signals that range from 0 to 1 V. What is the average digitization error in millivolts?
- *10-6. Define the following acronyms: RAM, ROM, PROM. Which is the most appropriate term to describe a computer chip that is altered during an on-site equipment upgrade?
- *10-7. A digitized radiograph is viewed on a 12-in.-wide monitor. If the actual width of the patient's chest is 14 in. and if the monitor resolution is set to 1024×1280 (height $\times$ width), does this monitor meet the ACR standard for use for primary diagnosis?
- *10-8. A $2048 \times 2048 \times 12$ -bit image is compressed at 10 to 1 compression ratio. How many bits are present in the compressed image?

*For problems marked with an asterisk, answers are provided on p. 492.

SUMMARY

- The number of **distinct values** N that may be recorded with n **bits** is given by the formula $N = 2^n$.
- **Bandwidth** is the maximum number of bits per second that may be transmitted over a communications channel.
 - Bandwidth in medical communications varies from thousands of bits per second (10^3 or **kbps**) in relatively inexpensive systems to billions of bits per second (10^9 or **Gbps**) in more expensive and specialized systems.
- Images may be thought of as three-dimensional data blocks, where a **matrix of pixels** is the “face” of the block and where the number of bits allocated for representation of the pixel value is the “depth” of the block.
- Medical images may be generally divided into **small matrix** and **large matrix** images.
 - **Small matrix images** (CT, MRI, ultrasound, nuclear medicine, digital fluorography) are 512×512 or smaller.
 - **Large matrix images** (CR, digitized film, direct digital capture x-ray images) are 1024×1024 or greater.

- In images, edges are generally associated with **higher spatial frequencies**, and contrast between large regions is generally associated with **lower spatial frequencies**.
- The Radiological Society of North America standard for official diagnostic interpretation of soft-copy images requires:
 - For **small matrix**:
 - Acquisition: $0.5 \text{ k} \times 0.5 \text{ k} \times 8$ bits or better
 - Display: $0.5 \text{ k} \times 0.48 \text{ k} \times 8$ bits or better
 - For **large matrix**:
 - Acquisition: 2.5 line pairs/mm and 1024 gray scale or better
 - Display: 2.5 line pairs/mm and 256 gray scale or better
- **Local image processing** involves movement of a small (e.g., 5×5 pixel) **kernel** over the image.
- **Global image processing** involves the application of a **filter function** or **operator** to the entire image.
- **High-pass filters** remove lower spatial frequencies while allowing higher spatial frequencies to remain.
- **Low-pass filters** remove higher spatial frequencies while allowing lower spatial frequencies to remain.
- Images may be **compressed** so that fewer bits are required for transmission or storage of the images.
 - In **loss-less compression**, the original image data block may be reconstructed to its original matrix size and depth with no error.
 - In **lossy compression**, some error is allowed in exchange for a reduction in the number of bits required to transmit or store the image.
- Picture Archiving and Communication Systems (**PACS**) consist of modules for acquisition, image processing, transmission, display and storage of images.
 - The internationally recognized image and data format for PACS systems is the Digital Imaging and Communications in Medicine (**DICOM**) standard.
- **Transmission media** include: copper wire ("twisted pair"), coaxial cable, optical fiber, and microwave carrier.
- **Transmission protocols** include: point-to-point presence (**PPP**), transmission control protocol/internet protocol (**TCP/IP**), and asynchronous transfer mode (**ATM**).
- The set of **file servers** and **routers** that are generally accessible throughout the world using a TCP/IP protocol is known as the **internet**.

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PROBABILITY AND STATISTICS

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■ OBJECTIVES

After studying this chapter, the reader will be able to:

- Distinguish between determinate and indeterminate errors, and give examples of each.
- Explain the concepts of

Accuracy	Mean
Precision	Median
Bias	Mode
True value	Standard deviation
Randomness	Standard error
Noise	Confidence limits

- Describe the properties of Poisson and normal (Gaussian) distributions.
- Delineate the rules for error propagation.
- Apply selected statistical tests:
 - Student’s *t*-test
 - Chi-squared test

■ INTRODUCTION

A random event is one in which the outcome is uncertain and can only be predicted by applicable rules of probability.

Statistics is the study of probability rules and their applications.

Analysis of error is a cornerstone of all research and clinical service. The presentation of patient data in the form of images involves uncertainty and requires statistical analysis. An appreciation of error allows the interpreter of images to judge the value of visual data and to develop more informed decisions about techniques of image acquisition. The choice of technique factors (e.g., kVp, mA, and time in radiography) influences (a) the appearance of statistical fluctuations (visual “noise”) in film images and (b) the random variations in pixel values in digital systems. Digital images are affected by the choice of pixel size, gray scale, and acquisition time. Trade-offs among these and other variables are dealt with continuously in diagnostic imaging. Nuclear medicine and ultrasonographic data are influenced by uncertainties that arise at all stages in the image-forming process. Data acquisition techniques in magnetic resonance imaging provide a wide variety of choices among variables such as pulse sequences, field strengths, and receiver coils, all of which affect the statistical quality of the images. For all these reasons, the imaging specialist must acquire a reasonable command of rudimentary statistical concepts.

■ NATURE OF ERROR

For every human problem, there is a solution that is simple, neat . . . and wrong. H. L. Mencken

The error of a measurement is the difference between the measured value and the “true value” for the variable being measured. Two categories of error exist, *determinate* and *indeterminate* errors. Determinate errors result from factors such as inadequate experimental design, malfunctioning equipment, and incomplete correction for extraneous influences. The influence of determinate errors upon experimental results can often be reduced by better instrumentation and thorough planning and execution of an experiment. For imaging systems, some examples of determinate errors include overexposure or underexposure to films in radiography, incorrect time-gain compensation (TGC) settings in ultrasonography, and a falloff in surface coil sensitivity in magnetic resonance images.

Determinate errors are sometimes called *systematic errors*; indeterminate errors are often termed *random errors*.

Indeterminate errors are errors that cannot be reduced by eliminating or correcting extraneous factors. They are caused by fundamental uncertainties in the variables being measured and in the processes of obtaining and displaying the measurements. In nuclear medicine, for example, the random nature of radioactive decay and the

statistical uncertainties in detecting ionizing radiations give rise to significant indeterminate errors.

The precision of a series of measurements describes the range or spread of the individual measurements from the average value of the series. Precision improves as the influence of indeterminate error is reduced. Precision does not describe the accuracy of a series of measurement. Accuracy is achieved only if the measured values agree with the true value determined by some method other than the measurement technique under consideration.

Usually, the “true value” for a measurement cannot be determined (if it could, the need for the measurement would disappear). However, a technique may be available that yields results that are generally accepted as being as close as possible to the true value. This technique and the result that it provides are referred to as a “gold standard.” An example of a gold standard is the use of biopsy results against which interpretations of diagnostic images are evaluated.

To increase the accuracy of experimental results, the influence of both indeterminate and determinate errors must be reduced. The contribution of determinate errors to a reduction in accuracy of a set of measurements is termed the bias of the measurements. The distinctions among precision, accuracy, and bias are illustrated in the margin.

PROBABILITY DISTRIBUTIONS

Indeterminate error implies that the results of a measurement may be different if the measurement is repeated. After many measurements have been recorded, a histogram can be constructed to show the number of observations for each measurement value (Figure 11-1). A histogram plotted as the frequency or fraction of times that each measurement is observed gives the probability that each specific value will be measured if the experiment is repeated. An accurate assessment of the probability of occurrence of each value would require an infinite number of measurements. If the sources of indeterminate error are known, however, a choice can be made among different theoretical probability distributions whose properties are depicted by mathematical analysis.

Poisson Distribution

Radioactive decay is a probabilistic (random) process. When many atoms are treated as a group, we may make statements such as “the number of atoms will be reduced to half of the initial number after a certain time (half-life) has elapsed.” With respect to a

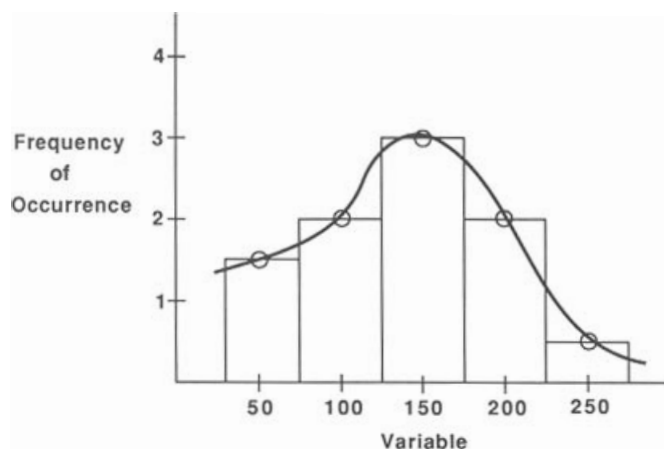
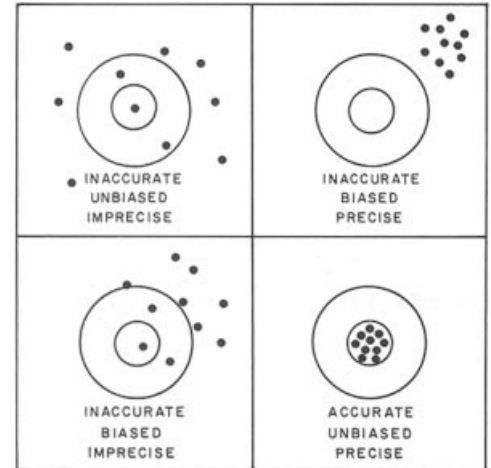


FIGURE 11-1

A histogram showing the frequency of occurrence of some variable. A histogram may be presented as bars, points, or a continuous line.

Precision: Is the consistency of repeated measurements of the same thing. Precision is sometimes referred to as *reliability*.

Accuracy is sometimes termed *validity*.



MARGIN FIGURE 11-1

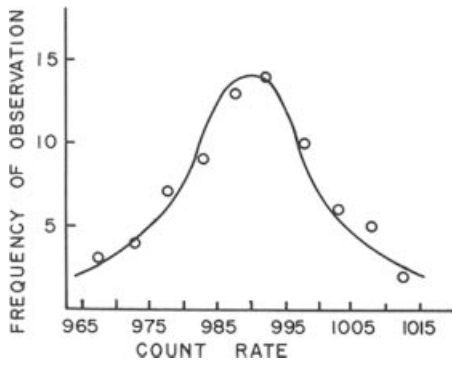
Precision, bias, and accuracy.

Bias is a systematic (i.e., not random) difference between the true value of a property and individual measurements of the property.

Bias is sometimes used incorrectly as a synonym for *prejudice*. Prejudice is a state of mind of an individual that creates a desire for a particular outcome.

A probability distribution reveals the likelihood of each of all possible outcomes of a set of measurements. The sum of all likelihoods (probabilities) (i.e., the area under a probability curve) is unity.

Simeon Denis Poisson (1781–1840) was a French mathematician who began studies as a physician but switched to mathematics after one of his patients died.

**MARGIN FIGURE 11-2**

Data for ^{137}Cs source that provided a count rate of about 1000 cpm. A count of 10,000 was accumulated for each measurement. The results of 100 measurements were plotted as a function of the number of times that the count rate fell within successive increments of 5 (i.e., between 975 and 980, 980 and 985, etc.). The Poisson curve of best fit is drawn through the data points (From F. Low.² Used with permission.)

A Poisson distribution is said to be a “discrete” distribution because it may contain only 0 or positive integers.

A Gaussian (normal) distribution is a bell-shaped symmetrical distribution characterized by two values: the mean of a set of measurements and the standard deviation of the measurements.

The Poisson distribution can be written with σ^2 in place of r .

Properties of a Gaussian distribution include the following:

- The distribution is continuous and symmetrical, with both tails extending to infinity.
- The arithmetic mean, median, and mode are identical.
- The likelihood of occurrence of a particular value depends on its distance from the mean value.
- The standard deviation is represented by the distance from the mean value to the inflection point of the distribution curve.

single atom, however, we cannot say with certainty when it will decay. The occurrence of radioactive decay (detection of “counts” in a radioactive sample) follows the Poisson probability distribution.¹ The shape of the Poisson probability distribution is given by the formula

$$p_n = \frac{r^n e^{-r}}{n!}$$

where p_n is the probability of obtaining a count n and r is the true average count for the sample. The term $n!$ [n -factorial] equals $[(n)(n-1)(n-2) \dots (2)(1)]$. The probability p_n is termed the Poisson probability density, and radioactive decay is said to follow a Poisson probability law. Because the true average count r (often called the true mean) for the sample cannot be determined, it is assumed to equal the average measured count (estimated mean), often referred to as the sample mean. Shown in the margin are the results of an experiment in which the count rate (number of counts measured over a given counting time) was measured repeatedly.

Example 11-1

What is the probability of obtaining a count of 12 when the average count is 15?

$$\begin{aligned} p_n &= \frac{r^n e^{-r}}{n!} \\ p_{12} &= \frac{(15)^{12} e^{-15}}{12!} \\ &= \frac{(129.7 \times 10^{12})(30.6 \times 10^{-8})}{(4.79 \times 10^8)} \\ &= 0.0829 \end{aligned} \quad (11-1)$$

The probability is 0.0829 (or 8.29%) that a count of 12 will be obtained when the average count is 15.

Normal Distribution

If a large number of counts is accumulated for a radioactive sample, then the Poisson distribution curve of frequency of occurrence versus counts is closely approximated by a Gaussian distribution curve. In this case the data are described reasonably well by the equation for a Gaussian (normal) probability density function. Illustrated in Figure 11-2 is the approach of a Poisson probability distribution to a Gaussian distribution. The equation for the Gaussian probability density g_n is

$$g_n = \frac{1}{\sqrt{2\pi r}} e^{-(n-r)^2/2r} \quad (11-2)$$

where g_n is the probability of observing a count n when the true count (estimated as the average count) is r .

Example 11-2

By using Eq. (11-2), estimate the probability of obtaining a count of 12 when the true or average count is 15.

$$\begin{aligned} g_n &= \frac{1}{\sqrt{2\pi r}} e^{-(n-r)^2/2r} \\ &= \frac{1}{\sqrt{2\pi}(15)} e^{-(12-15)^2/2(15)} \\ &= 0.103e^{-0.3} \\ &= 0.0764 \end{aligned} \quad (11-3)$$

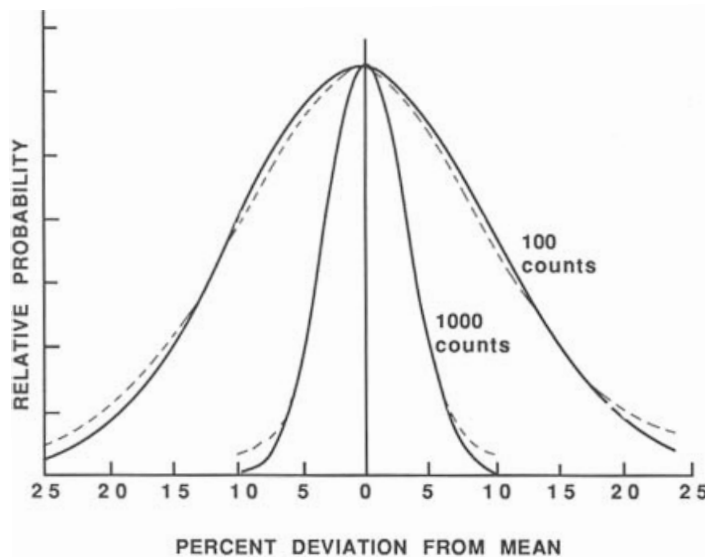


FIGURE 11-2

Probability of observing a particular count is plotted for 100 measurements and 1000 measurements. *Dashed curves* illustrate Gaussian (normal) probability density functions, and *solid curves* illustrate Poisson probability density functions. Curves have been normalized to equal heights at the mean.

The probability is 0.0764 (or 7.64%) that a count of 12 will be obtained when the average count is 15, estimated by assuming that radioactive decay follows a Gaussian rather than a Poisson probability distribution.

■ SIGNAL AND NOISE

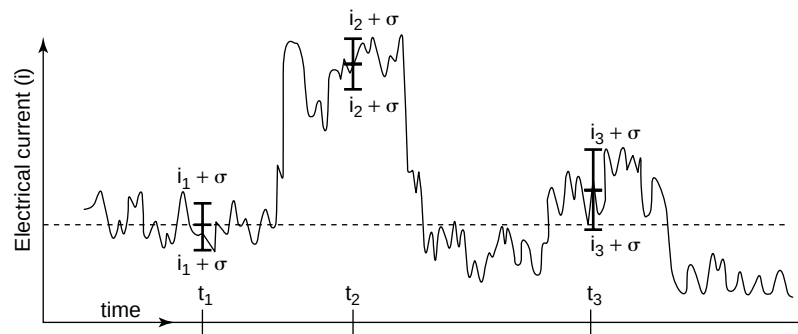
Many of the challenges of imaging systems may be reduced to the simple problem of detecting a signal against a background of noise. The principle of signal to noise can be illustrated by examining the electrical signal in a wire. The current in a wire plotted as a function of time yields a graph such as shown in Figure 11-3 where there is a fluctuation, or noise σ , at all times. The current i is measured at three different times t_1 , t_2 , and t_3 . The current i_0 is the average “background level.” The problem of signal detection may then be stated as a question: Are the currents i_1 , i_2 , and i_3 measured at times t_1 , t_2 , and t_3 significantly different from the background level i_0 ? If so, then they can be interpreted as signals occurring at times t_1 , t_2 , and t_3 . If not, they should be interpreted as statistical fluctuations in the background current and ignored. Unfortunately, there are no definitive answers to these questions. If the noise σ is truly random, we cannot say with absolute certainty that the currents i_1 , i_2 , and i_3 are signals rather than simply fluctuations in the background current, no matter how different the levels may be.

There are two approaches to this dilemma. The first approach is to state the probability that currents i_1 , i_2 , and i_3 are different from the background i_0 . This leaves the final decision up to the reader. However, there are levels of probability that are commonly accepted as sufficient to say that two signals are significantly different. For example, one might say that there is a 95% chance that a peak such as i_2 is not a random fluctuation in the background current but instead represents a true signal superimposed on the background current. This approach is the foundation of the statistical t -test discussed later in this chapter.

The second approach to the separation of signal and background is to determine beforehand that a multiple of noise or background will be accepted as a signal. For

A series of events is said to be *random* if every event has an equal probability of occurring. “White noise” refers to the presence of background signals that are random in nature.

The background signals that constitute white noise are “independent events” which have the property that each event has no influence on the likelihood or value of a previous or subsequent event.

**FIGURE 11-3**

The problem of signal detection against background noise. Electrical currents i_1 , i_2 , and i_3 are measured at three times, t_1 , t_2 , and t_3 . For each measurement there is a mean value i and a standard deviation σ . A current i_0 is assumed to be the background value. That is, i_0 was measured at a time when, through prior information, it was known to be unlikely that there would be a signal (a current significantly above background level and noise) resulting in our estimate of background level (i_0) and noise (σ). The problem is to determine if there is a “signal” at time t_1 , t_2 or t_3 . The central problem of signal detection is to determine the current i that is required to be accepted as a signal that is distinct from background noise. In the example shown here, it appears that there is a clear signal (i.e., distinct from the noise) at t_2 , while the signals at t_1 and t_3 are not strong enough to “rise above the noise.” Statistical techniques, such as specialized versions of the t -test, may be used to determine the probability that it was a “real” signal.

signal i_2 we define (with reference to Figure 11-3)

$$\text{Signal-to-background ratio} = \frac{i_2}{i_0}$$

$$\text{Signal-to-noise ratio} = \frac{i_2}{\sigma}$$

where the noise σ is a measure of the fluctuation around the background level i_0 . In diagnostic imaging, a value of the ratio of 3 to 5 often is accepted as indicative of a signal.

The problem of signal detection against background noise can be extended to three dimensions, as shown in Figure 11-4 (in this image, spatial coordinates are plotted for two dimensions, vertical and horizontal, and gray scale is displayed as the third dimension). In a Poisson distribution, noise varies with the average value of a series of measurements. This relationship is assumed in many applications of digital imaging where the noise (standard deviation) associated with a pixel value is assumed to equal the square root of the pixel value.

METHODS TO DESCRIBE PROBABILITY DISTRIBUTIONS

Mean, Median, and Mode

When describing the results of several measurements, a single value that summarizes the results is useful. For example, consider the data in Table 11-1. A typical approach to the problem is to average the data. However, the term “average value” is nonspecific. Exact reporting requires use of one of the following three terms: mean, median, or mode.

Mean

The mean is the result obtained by summing (adding together) all measured values and dividing by the number of values. The sum of the nine values in Table 11-1 is 39. Therefore, the sample mean is $39/9 = 4.3$.

The mean is the arithmetic average of measured values. Usually, the mean is intended when the term “average” or “average value” is used.

The median is also known as the 50th percentile.

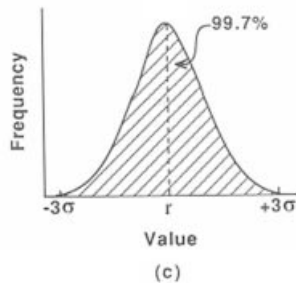
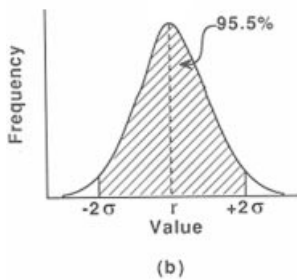
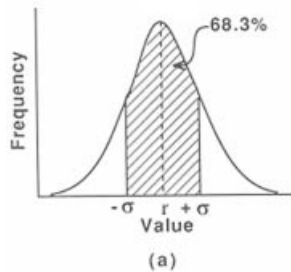
The mode is the most common value in a set of measurements.

If there were an even number of values, the median would be the average of the two values on either side of a hypothetical center value.

Mode

The mode is the value that appears most often in a set of data. When the data in Table 11-1 are placed in order of frequency of occurrence, the value 6 occurs most often. Hence, 6 is the mode for this series of measurements.

Because the mean, median, and mode may all be different, it is important to specify the type of “average” that is used. For example, suppose that the data in Table 11-1 are the scores obtained by radiologists diagnosing a disorder with a new type of imaging system. In these data a score of 6 represents perfect agreement with biopsy results, and a score of 0 represents complete disagreement. We may compare the mean scores for other imaging systems against the mean score of 4.3 for the new system. The mode, or most frequently reported score, is the high score of 6. Perhaps the experimental setup should be examined to see whether the viewing conditions, training, experience, or other factors were more favorable for the four radiologists who scored 6. The 50th percentile score (median) was 4, and the next highest score was the maximum. This suggests that the radiologists can be separated into two groups, one whose performance with the new system is excellent and another whose performance is inferior. This conclusion suggests that any differences between the two groups (e.g., training, experience, visual acuity, viewing conditions, etc.) might explain the difference in performance. Of course, one possibility is that the sample size (nine radiologists) is too small to represent radiologists in general.



MARGIN FIGURE 11-3

The three Gaussian probability distributions shown above are centered about the mean value r .

A: The area defined by ± 1 SD is 68.3% of the total area under the Gaussian. **B:** Two standard deviations on either side of the mean encompass 95.5% of the total area. **C:** Three standard deviations encompass 99.7%.

In a Gaussian distribution, one σ on either side of the mean define the inflection points of the distribution.

Standard Deviation

The standard deviation for a series of measurements describes the precision or reproducibility of the measurements. If a series of measurements is distributed according to the Poisson probability distribution, the standard deviation σ is given by

$$\sigma = \sqrt{r_t}$$

where r_t is the true mean (value) for the series. If the true mean is estimated as the mean value r for a series of measurements, then the estimated standard deviation for the measurements is

$$\sigma = \sqrt{r}$$

If only one measured value, n , is available, then the standard deviation may be estimated by assuming that the value n represents the true mean.

In a normal distribution, 68.3% of all measured values fall within 1 σ on either side of the mean, 95.5% within 2 σ , and 99.7% within limits of 3 σ . If an experiment yields results that follow a normal distribution, there is a 68.3% likelihood that any particular result will fall within 1 σ of the mean, a 95.5% chance that the result will be within 2 σ , and a 99.7% chance that the result will be within 3 σ of the mean.

Three “types” of standard deviations are commonly cited. These types are described briefly below.

Standard Deviation of True Distribution

This is the value that we would like to report for an experiment but can never actually achieve because an infinite number of measurements would be required.

Standard Deviation Estimated from a Sample Mean

If a series of measurements is obtained and the mean value determined, the standard deviation can be estimated by taking the square root of the estimated mean. An extreme case of this procedure occurs when a single measurement is obtained of a variable assumed to follow a Poisson distribution. In this case, we might assume

that the measured value is the mean of the distribution. We then report the single measurement as the mean, and we report the square root of the single measurement as the standard deviation. The number of samples actually taken should always be reported when a standard deviation is estimated from a mean. The rather inexact technique of estimating the standard deviation from a single measurement is often necessary in digital imaging techniques where the “noise” of a pixel value is estimated as the square root of the pixel value. When a value is measured only once, this inexact estimate is the only choice available.

Standard Deviation of a Series of Measurements

From a series of measurements the quantity known as the sample standard deviation can be constructed. The sample standard deviation σ_s is

$$\sigma_s = \sqrt{\frac{\sum_{i=1}^N (x_i - r)^2}{N - 1}} \quad (11-4)$$

where N measurements of a variable are made. The individual measurements yield the values x_i , where i takes on the values 1, 2, 3, ..., N . The value x_1 is the first measurement, x_2 is the second measurement, and so on. The mean value of the sample is r . The quantity

$$\frac{\sum_{i=1}^N (x_i - r)^2}{N - 1}$$

is known as the variance of the measurements. That is, the variance is σ_s^2 , the square of the sample standard deviation.

If the formula for the sample standard deviation were applied to a “true” Poisson population (this can only be done mathematically since a “true” population or data set requires an infinite amount of data), it would reveal that the standard deviation equals the square root of the mean value. For any actual series of measurements, the sample standard deviation may yield a number that is “close” to the square root of the sample mean. The closer the sample distribution approximates a true Poisson distribution, the more nearly the sample standard deviation equals the square root of the sample mean.

A Poisson distribution has the following properties:

- It is composed of discontinuous discrete data;
- A large number of measurements is performed;
- The likelihood of any particular result is small;
- The chance of any particular measurement is independent of measurements that preceded it.

Example 11-3

Determine the standard deviation for the set of measurements given in Table 11-2:

$$\begin{aligned} \text{Mean} = r &= 43,950 \\ \sum (x_i - r)^2 &= 1,302,400 \\ \sigma_s &= \sqrt{\frac{\sum (x_i - r)^2}{N - 1}} \\ &= \sqrt{\frac{1,302,400}{10 - 1}} \\ \sigma_s &= 380 \end{aligned}$$

The mean may be expressed as $43,950 \pm 380$, where 380 is understood to be 1σ .

Assume that the data in Table 11-2 are from a population having a Poisson distribution, and estimate the standard deviation from the mean value alone.

$$\begin{aligned} \sigma \sqrt{r} &= \sqrt{43,950} \\ &= 210 \end{aligned}$$

TABLE 11-2 Data for Computation of Mean and Standard Deviation

x_i (Measurement)	$x_i - r$ (Deviation)	$(x_i - r)^2$
43,440	-510	260,100
43,720	-230	52,900
44,130	180	32,400
43,760	-190	36,100
44,390	440	193,600
43,810	-140	19,600
44,740	790	624,100
43,750	-200	40,000
44,010	60	3,600
43,750	-200	40,000
mean = $\frac{\sum x_i}{N} = 43,950$		$1,302,400 = \sum (x_i - r)^2$

Note that this value differs from the sample standard deviation of 380 derived above. This result reveals that the data are distributed more widely than would be predicted for a Poisson distribution having the same mean value.

■ PROPAGATION OF ERROR

Variables should be managed according to appropriate statistical models. When variables are modified by operations such as scaling (multiplication or division by a constant), statistical parameters such as the standard deviation must be treated appropriately. For example, counts measured in a nuclear decay experiment have a Poisson distribution, and it may be assumed that the standard deviation is equal to the square root of the mean. However, a count rate obtained by dividing the number of counts by the counting time must be handled differently. If all of the data values x_i in Eq. (11-4) are divided by the counting time t_1 , then the sample mean r is also divided by t , and the standard deviation of a count rate σ_c is

$$\begin{aligned}
 \sigma_c &= \sqrt{\frac{\sum_{i=1}^N \left(\frac{x_i}{t} - \frac{r}{t}\right)^2}{N-1}} \\
 &= \sqrt{\frac{1}{t^2} \frac{\sum_{i=1}^N (x_i - r)^2}{N-1}} \\
 &= \frac{1}{t} \sqrt{\frac{\sum_{i=1}^N (x_i - r)^2}{N-1}} \\
 &= \frac{1}{t} \sigma_s
 \end{aligned}$$

The estimated standard deviation σ_c of a count rate is then

$$\sigma_c = \frac{\sigma_s}{t}$$

where σ_s is the estimated standard deviation of a count n , and the average count rate is determined by dividing the count n by the counting time t . Because

$$\sigma_s = \sqrt{n}$$

we have

$$\sigma_c = \frac{\sqrt{n}}{t}$$

In estimating the precision of a count rate, the count is assumed to be the product of a random process whereas the time over which the count is measured is assumed to be known exactly.

TABLE 11-3 Results of Arithmetic Operations with Numbers A and B ^a

Arithmetic Operation	First Number	Second Number	Result $\pm$ Standard Deviation
Addition	$(A \pm \sigma_A)$	$+(B \pm \sigma_B)$	$(A + B) \pm \sqrt{\sigma_A^2 + \sigma_B^2}$
Subtraction	$(A \pm \sigma_A)$	$-(B \pm \sigma_B)$	$(A - B) \pm \sqrt{\sigma_A^2 + \sigma_B^2}$
Multiplication	$(A \pm \sigma_A)$	$\times(B \pm \sigma_B)$	$(AB)[1 \pm \sqrt{(\sigma_A/A)^2 + (\sigma_B/B)^2}]$
Division	$(A \pm \sigma_A)$	$\div(B \pm \sigma_B)$	$(A/B)[1 \pm \sqrt{(\sigma_A/A)^2 + (\sigma_B/B)^2}]$
Scaling	$C(A \pm \sigma_A)$		$CA \pm C\sigma_A$

^aThe precision of A and B is described by the standard deviations σ_A and σ_B . C is a constant.

The count n is the average count rate c multiplied by the counting time t . Therefore

$$\begin{aligned}\sigma_c &= \frac{\sqrt{ct}}{t} \\ &= \sqrt{\frac{c}{t}}\end{aligned}\quad (11-5)$$

Hence the estimated standard deviation of a count rate (average or instantaneous) is determined by first dividing the count rate by the counting time and then taking the square root of the result.

If the precision of numbers A and B is denoted by standard deviations σ_A and σ_B , then the standard deviations of the results of arithmetic operations involving A and B may be computed with the expressions in Table 11-3.

A frequent mistake is to assume that the standard deviation of a count rate is the square root of the count rate c , that is, $\sigma_c \neq \sqrt{c}$.

Example 11-4

What is the net count and standard deviation for a sample from a nuclear counting experiment if the count is 400 ± 22 for the sample plus background and the background count is 64 ± 10 ?

$$\begin{aligned}\text{Sample net count} &= (A - B) \pm \sqrt{\sigma_A^2 + \sigma_B^2} \\ &= (400 - 64) \pm \sqrt{(22)^2 + (10)^2} \\ &= 336 \pm 24 \text{ counts}\end{aligned}$$

Example 11-5

From repeated counts of a radioactive sample, the mean sample plus background count was 232 ± 16 . The counting time for each count was 2.0 ± 0.1 minute. What are the mean (sample plus background) count rate and the estimated standard deviation of the mean count rate?

$$\begin{aligned}\text{Mean sample plus background count rate} &= \frac{\text{Mean (sample plus background)}}{\text{Time}} \\ &= \left(\frac{A}{B}\right) \left[1 \pm \sqrt{\left(\frac{\sigma_A}{A}\right)^2 + \left(\frac{\sigma_B}{B}\right)^2}\right] \\ &= \left(\frac{232}{2.0}\right) \left[1 \pm \sqrt{\left(\frac{16}{232}\right)^2 + \left(\frac{0.1}{2.0}\right)^2}\right] \\ &= 116 \pm 10 \text{ cpm}\end{aligned}$$

Example 11-6

A 10-minute count of sample plus background yields 10,000 counts, and a 6-minute count of background alone yields 1296 counts. What are the net sample count rate and its standard deviation?

$$\text{Sample plus background count rate} = \frac{10,000 \text{ counts}}{10 \text{ min}} = 1000 \text{ cpm}$$

$$\text{Background count rate} = \frac{1296 \text{ counts}}{6 \text{ min}} = 216 \text{ cpm}$$

$$\begin{aligned} \sigma_C \text{ for sample plus background count rate} &= \sqrt{\frac{c}{t}} = \sqrt{\frac{1000 \text{ cpm}}{10 \text{ min}}} = \sqrt{100} \\ &= 10 \text{ cpm} \end{aligned}$$

$$\sigma_C \text{ for background count rate} = \sqrt{\frac{c}{t}} = \sqrt{\frac{216 \text{ cpm}}{6 \text{ min}}} = \sqrt{36} = 6 \text{ cpm}$$

$$\begin{aligned} \text{Net sample count rate} &= (A - B) \pm \sqrt{\sigma_A^2 + \sigma_B^2} \\ &= (1000 - 216) \pm \sqrt{(10)^2 + (6)^2} \\ &= 784 \pm \sqrt{136} \\ &= 784 \pm 12 \end{aligned}$$

■ OTHER METHODS FOR DESCRIBING PRECISION

In addition to the standard deviation, a number of other methods can be used to describe the precision of data. Only a brief explanation of these methods is provided here, and the reader is referred to standard texts on probability and statistics for a more complete explanation.³⁻⁵

Probable error (PE) may be used in place of the standard deviation to describe the precision of data. The probability is 0.5 that any particular measurement will differ from the true mean by an amount greater than the PE. The probable error $PE = 0.6745\sigma$.

In place of the standard deviation, 9/10 error may be used to describe the precision of data. The probability is 0.1 that any particular measurement will differ from the true mean by an amount greater than the 9/10 error.

A 95/100 error is similar to the 9/10 error, except that the probability is 0.05 that the difference between a particular measurement and the true mean is greater than the 95/100 error.

Confidence limits may be used to estimate the precision of a number when repeated measurements are not made. For example, if a result is stated as $A \pm a$, where $\pm a$ defines the limits of the 95% confidence interval, then it may be said with "95% confidence" that a second measurement of A will fall between $A - a$ and $A + a$. For a normal distribution, the 95% confidence interval is approximately $A \pm 2\sigma$.

Confidence limits are also referred to as a confidence interval, with the limits serving as the lower and upper bounds of the interval. Although 95% and 99% confidence intervals are frequently encountered, an interval of any magnitude could be used.

Fractional Standard Deviation

The fractional standard deviation (σ/r) may be used in place of the standard deviation to describe the precision of data. If r represents the mean, then $\sigma = \sqrt{r}$, and the fractional standard deviation (σ/r) is $1/\sqrt{r}$. For example, σ for the number 10,000 is 100, and the fractional standard deviation is $100/10,000 = 0.01$. The percent standard deviation ($\%\sigma$) is the fractional standard deviation multiplied by 100.

In the example above, $\% \sigma = 1$. The percent standard deviation of a count rate ($\% \sigma_c$) is

$$\% \sigma_c = 100 \sqrt{\frac{1}{ct}} \quad (11-6)$$

where c is the average count rate measured over a time t .

Example 11-7

For a count rate of 1000 cpm, what counting time is necessary to achieve a percent standard deviation of 1%?

$$\begin{aligned} \% \sigma_c &= 100 \sqrt{\frac{1}{ct}} \\ 1\% &= 100 \sqrt{\frac{1}{(1000 \text{ cpm})t}} \\ 1000t &= (100)^2 = 10,000 \\ t &= 10 \text{ min} \end{aligned}$$

Estimated percent standard deviations for different values are as follows:

Value	$\% \sigma$
10	32
100	10
1000	3.2
10,000	1
100,000	0.32
1,000,000	0.1

Standard Deviation of the Mean

The standard deviation of the mean σ_m yields an estimate of the precision of the mean of a set of measurements. For N individual measurements constituting a mean, $\sigma_M = \sigma_s / \sqrt{N}$, where σ_s is the standard deviation of the individual measurements.

The standard deviation of the mean is sometimes referred to as the *standard error*.

Example 11-8

Determine the standard deviation of the mean σ_M for the data in Table 11-2.

For these data,

$$\begin{aligned} \sigma_s &= 380 \text{ (Example 11-3)} \\ N &= 10 \end{aligned}$$

Therefore

$$\sigma_M = \frac{\sigma}{\sqrt{N}} = \frac{380}{3.16} = 120$$

The standard deviation of the mean (120) always is smaller than the sample standard deviation.

The standard deviation of the mean is a measure of the spread of mean values expected when sets of data (such as the 10 measurements in Table 11-2) are repeated several times. If the experiment were repeated once, there is a 68% chance that the mean value of the next data set would fall within ± 1 standard deviation of the mean of the first set.

The standard deviation of the mean should not be confused with the standard deviation of a set of data. It is misleading to report the standard deviation of the mean instead of the standard deviation of individual values of a set of data unless it has been made clear that the spread of the *mean* values is the parameter of interest. It is usually the range of individual measurements that is of interest. Under such circumstances, reporting the standard deviation of the mean simply makes the data look more reproducible than they actually are.

Two factors influence the value of σ_M : (1) the variability of the original set of measurements, as represented by σ ; and (2) the size of the sample N .

■ SELECTED STATISTICAL TESTS

A number of statistical tests may be applied to a set of measurements. A few of these tests are described briefly in this section.

Student's *t*-Test

The *t*-test was published in 1908 by W. S. Gosset, a brewer who worked for the Guinness Company. Gosset used the pseudonym “Student” in the publication, and the test became known as Student’s *t*-test.

Student’s *t*-test is a method for testing for the significance of the difference between two measurements, or two sets of measurements. The *t*-value for the difference between two measurements n_1 and n_2 is

$$t\text{-value} = \frac{|n_1 - n_2|}{\sqrt{\sigma_1^2 + \sigma_2^2}} \tag{11-7}$$

where σ_1 and σ_2 are the standard deviations for the measurements n_1 and n_2 , and the vertical bars enclosing n_1 and n_2 indicate that the absolute or positive value of the difference is used for the computation. With the computed *t*-value and Table 11-4, the probability *p* may be determined that the difference in the numbers is simply statistical in nature and is not a real difference between dissimilar samples.

The probability *p* is the probability of a deviation from a true value equal to or greater than the measured value, if the Null Hypothesis is true (i.e., the deviation is simply statistical and not real).

Example 11-9

Activity in livers excised from two rabbits given ^{99m}TcS intravenously was counted for 1 minute. The net count rate was 1524 ± 47 cpm for the first liver and

TABLE 11-4 Cumulative Normal Frequency Distribution

<i>t</i> -Value	<i>p</i>	<i>t</i> -Value	<i>p</i>
0.0	1.000	2.5	0.0124
0.1	0.920	2.6	0.0093
0.2	0.841	2.7	0.0069
0.3	0.764	2.8	0.0051
0.4	0.689	2.9	0.0037
0.5	0.617	3.0	0.00270
0.6	0.548	3.1	0.00194
0.7	0.483	3.2	0.00136
0.8	0.423	3.3	0.00096
0.9	0.368	3.4	0.00068
1.0	0.317	3.5	0.00046
1.1	0.272	3.6	0.00032
1.2	0.230	3.7	0.00022
1.3	0.194	3.8	0.00014
1.4	0.162	3.9	0.00010
1.5	0.134	4.0	0.0000634
1.6	0.110	4.1	0.0000414
1.7	0.090	4.2	0.0000266
1.8	0.072	4.3	0.0000170
1.9	0.060	4.4	0.0000108
2.0	0.046	4.5	0.0000068
2.1	0.036	4.6	0.0000042
2.2	0.028	4.7	0.0000026
2.3	0.022	4.8	0.0000016
2.4	0.016	4.9	0.0000010

1601 $\pm$ 49 cpm for the second. Is the difference significant?

$$\begin{aligned}
 t\text{-value} &= \frac{|n_1 - n_2|}{\sqrt{\sigma_1^2 + \sigma_2^2}} \\
 &= \frac{|1,524 - 1,601|}{\sqrt{(47)^2 + (49)^2}} \\
 &= \frac{77}{68} \\
 &= 1.13
 \end{aligned}$$

From Table 11-4, the probability p is 0.259 (or 25.9%) that the difference is attributable to random variation of the count rate for similar samples. The probability is $1 - 0.259 = 0.741$ that the difference between the samples is significant.

Values of $p > 0.01$ seldom are considered indicative of a significant difference in values between measurements.

Chi-Square (χ^2) Test

The chi-square test is used to determine the goodness of fit of measured data to a probability density function. From a series of repeated measures, the value of χ^2 may be computed:

$$\chi^2 = \frac{1}{r} \sum_{i=1}^N (x_i - r)^2 \quad (11-8)$$

where x_i represents each of N individual measurements and r is the sample mean. The formula for computing χ^2 from a sample resembles the formula for calculating the sample standard deviation [Eq. (11-4)]. A little algebra shows that the two are related by the equation

$$\chi^2 = \frac{(N-1)\sigma_s^2}{r} \quad (11-9)$$

If the variable we are measuring has a Poisson distribution, then the sample standard deviation σ_s should be close to the square root of the mean r of the sample. That is, the square of the sample standard deviation (the variance) should equal the sample mean. If that is true, then Eq. (11-9) shows that χ^2 is equal to $N - 1$, or one less than the number of data points.

χ^2 may be used as a measure of how closely the sample data correspond to a Poisson distribution. From the computed value of χ^2 , the number of degrees of freedom ($N - 1$), and data in Table 11-5, a probability p may be determined. Values of p less than 0.05 suggest that the data are distributed over a wider range of values than would be expected for a Poisson distribution. Values of p greater than 0.95 suggest that the data are confined to a smaller range of values than that predicted for a Poisson distribution.

Example 11-10

A series of 25 measurements on a radioactive sample provides a mean of 950 and a value of 17,526 for

$$\sum_{i=1}^N (x_i - r)^2$$

The Null Hypothesis is the assumption that a measured value represents the true value and that any deviation between the two is purely statistical and not real. The Null Hypothesis gives rise to two types of error:

- **Type I Error:** The Null Hypothesis is rejected as false even though it is true;
- **Type II Error:** The Null Hypothesis is accepted as true even though it is false.

The number of degrees of freedom is the number of observations that are free to vary without restriction. For example, there are $(N - 1)$ degrees of freedom in a box of N chocolates, because after $(N - 1)$ chocolates have been selected, there is no freedom in the selection of the last chocolate.

TABLE 11-5 Chi-Square Values

Degrees of Freedom ($N - 1$)	There Is a Probability of						
	0.99	0.95	0.90	0.50	0.10	0.05	0.01
	That the Calculated Value of Chi-Square Will Be Equal to or Greater Than						
2	0.020	0.103	0.211	1.386	4.605	5.991	9.210
3	0.115	0.352	0.584	2.366	6.251	7.815	11.345
4	0.297	0.711	1.064	3.357	7.779	9.488	13.277
5	0.554	1.145	1.610	4.351	9.236	11.070	15.086
6	0.872	1.635	2.204	5.348	10.645	12.592	16.812
7	1.239	2.167	2.833	6.346	12.017	14.067	18.475
8	1.646	2.733	3.490	7.344	13.362	15.507	20.090
9	2.088	3.325	4.168	8.343	14.684	16.919	21.666
10	2.558	3.940	4.865	9.342	15.987	18.307	23.209
11	3.053	4.575	5.578	10.341	17.275	19.675	24.725
12	3.571	5.226	6.304	11.340	18.549	21.026	26.217
	4.107	5.892	7.042	12.340	19.812	22.362	27.688
	4.660	6.571	7.790	13.339	21.064	23.685	29.141
	5.229	7.261	8.547	14.339	22.307	24.996	30.578
	5.812	7.962	9.312	15.338	23.542	26.296	32.000
	6.408	8.672	10.085	16.338	24.769	27.587	33.409
	7.015	9.390	10.865	17.338	25.989	28.869	34.805
	7.633	10.117	11.651	18.338	27.204	30.144	36.191
	8.260	10.851	12.443	19.337	28.412	31.410	37.566
	8.897	11.591	13.240	20.337	29.615	32.671	38.932
	9.542	12.338	14.041	21.337	30.813	33.924	40.289
	10.196	13.091	14.848	22.337	32.007	35.172	41.638
	10.856	13.848	15.659	23.337	33.196	36.415	42.980
	11.534	14.611	16.473	24.337	34.382	37.566	44.314
	12.198	15.379	17.292	25.336	35.563	38.885	45.642
	12.879	16.151	18.114	26.336	36.741	40.113	46.963
	13.565	16.928	18.939	27.336	37.916	41.337	48.278
	14.256	17.708	19.768	28.336	39.087	42.557	49.588

Do the data appear to follow a normal distribution?

$$\begin{aligned}
 \chi^2 &= \frac{1}{r} \sum_{i=1}^N (x_i - r)^2 \\
 &= \frac{1}{950} (17,526) \\
 &= 18.5
 \end{aligned}$$

The χ^2 test is often used in the classroom to demonstrate that the data from Gregor Mendel's famous experiments with peas, which formed the basis of his theory of inheritance, were probably falsified.^{6,7}

With $N - 1$, or 24 degrees of freedom, this value for χ^2 falls between probabilities 0.9 and 0.5 and suggests that the data are distributed as a Poisson data set.

Example 11-11

Determine whether the data in Table 11-2 are distributed according to a Poisson distribution.

From Table 11-2,

$$\begin{aligned}
 \sum_{i=1}^N (x_i - r)^2 &= 1,302,400 \\
 \chi^2 &= \frac{1}{r} \sum_{i=1}^N (x_i - r)^2 = \frac{1,302,400}{43,950} = 29.6
 \end{aligned}$$

With $N - 1$, or 9 degrees of freedom, this value of χ^2 falls below a p value of 0.01 and suggests that the data are not distributed according to a Poisson distribution.

SUMMARY

- Errors can be categorized as determinate (systematic) and indeterminate (random).
- A series of measurements may be characterized in terms of its accuracy, precision, and bias.
- Common probability distributions are the
 - Poisson distribution
 - Normal (Gaussian) distribution
- The mean, median, and mode are single values that can be determined for any probability distribution; the values are identical for a normal distribution.
- The precision of a series of measurements is described by the standard deviation (sometimes expressed as the fractional standard deviation).
- Mathematical manipulations of imprecise data lead to the propagation of errors that can be described analytically for a normal distribution.
- Student's t -test is a method for determining the statistical significance of a difference between sets of measurements.
- The chi-square test is used to determine the "goodness of fit" of measured data to an expected probability distribution.

PROBLEMS

- *11-1. Suppose 3600 counts were accumulated for a sample counted for 10 minutes. Background contributed 784 counts over a counting interval of 6 minutes.
- Find the estimated standard deviation and the percent estimated standard deviation of the sample count rate uncorrected for background.
 - What are the estimated standard deviation and the percent estimated standard deviation for the background count rate?
 - What are the estimated standard deviation and the percent estimated standard deviation of the sample count rate corrected for background?
- 11-2. A medical researcher carefully measures the length of the fetal humerus in 50 ultrasound studies by using electronically generated distance-measuring "calipers" on the display. The researcher obtains a mean value of 5.5 cm with a standard deviation of 2 mm. After the measurements have been obtained, the researcher discovers through studies of excised bone that the electronically generated caliper measurements are consistently 2% larger than the true length.
- Describe the researcher's original results in terms of bias, precision, and accuracy.
 - How can the mean and standard deviation of the original data be corrected?
- *11-3. A 2-minute count of 3310 was measured for a radioactive sample. A second sample furnished 4709 counts in 3 minutes. Is the difference in count rate between the two samples significant?
- 11-4. Show that the formula for [Eq. (11-8)] can be rewritten in terms of the formula for the sample standard deviation [Eq. (11-4)] to yield Eq. (11-9).
- 11-5. A series of radioactive sample of short half-life will be counted one per day over a period of several days. Explain how variations in the observed counts caused by fluctuations in the sensitivity of the counting system can be compensated.

*For those problems marked with an asterisk, answers are provided on p. 492.

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■ OBJECTIVES

By studying this chapter, the reader should be able to:

- Describe the principles and application of rate measurements with radioactive nuclides.
- Compute volumes using the dilution method.
- Identify the function of each component of a scintillation camera.
- Explain the properties and uses of different collimators.
- List the advantages of a single-crystal scintillation camera.
- Trace the progression of imaging data in a single-crystal camera from release of a γ ray in the patient to display of a bright spot on the display.
- Explain the origin and correction of the resolving time of a scintillation camera.
- Discuss the acquisition, presentation and properties of images for single-photon emission tomography.
- Discuss the acquisition, presentation and properties of images for positron emission tomography.
- Document recent advances in positron emission tomography (PET).

Nuclear diagnostic studies can be categorized into four general groups:

- Localization
- Dilution
- Flow or diffusion
- Biochemical and metabolic properties

Many of these properties were first demonstrated in the 1920s by George de Hevesy, a Hungarian physicist working in Denmark. One of his more well-known experiments was use of a radioactive tracer to prove that the hash served at his boarding house contained meat he had left on his plate the evening before.

In accepting the Nobel Prize in Chemistry in 1943, de Hevesy stated: "The most remarkable result obtained in the study of the application of isotope indicators is perhaps the discovery of the dynamic state of the body constituents. The molecules building up the plant or animal organism are incessantly renewed."

Before the development of the scintillation detector, thyroid uptakes were measured with a collimated Geiger-Müller counter.

In the 1930s and 1940s, thyroid function was studied with a variety of radioactive isotopes of iodine, including ^{126}I , ^{128}I , ^{130}I and ^{131}I . After World War II, ^{131}I became the nuclide of choice because it could be produced in copious amounts in nuclear reactors. More recently, ^{123}I has replaced ^{131}I as the preferred nuclide for thyroid uptake measurements.

■ INTRODUCTION

Most nuclear medicine studies require the use of a detector outside the body to measure the rate of accumulation, release, or distribution of radioactivity in a particular region inside the body. The rate of accumulation or release of radioactivity may be measured with one or more detectors, usually NaI(Tl) scintillation crystals, positioned at fixed locations outside the body. Images of the distribution of radioactivity usually are obtained with a stationary imaging device known as a scintillation camera.

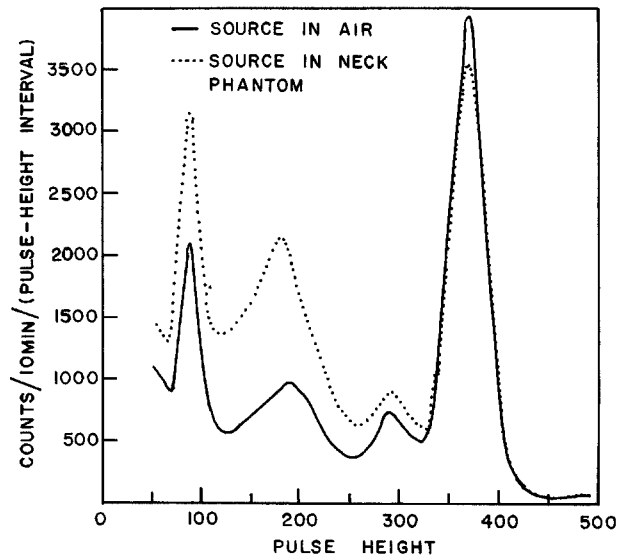
■ MEASUREMENT OF ACCUMULATION AND EXCRETION RATES

Measurement of the accumulation of radioactive iodine in the thyroid gland was the first routine clinical use of a radioactive nuclide. Since early tests of thyroid function with radioactive iodine, many diagnostic applications of radioactive nuclides have been developed. Many of these applications require measurement of the change in radioactivity in a selected region of the body as a function of time. Some of the applications are discussed briefly in this section.

Thyroid Uptake

The rate of accumulation (uptake) of iodine in the thyroid gland may be estimated from measurements of the radioactivity in the thyroid at specified times (usually 6 and 24 hours) after a selected amount of radioactive iodine (e.g., ~ 10 MBq of ^{123}I) has been administered orally. Radioactivity in the thyroid is usually measured with a NaI(Tl) scintillation crystal at least 3 cm in diameter by 3 cm thick that is positioned 20 to 30 cm in front of the patient and in line with the isthmus of the thyroid. The *field of view* of the detector is the area perpendicular to the extended axis of the detector at the surface of the patient over which the count rate remains above 90% of the count rate at the center of the field. The detector should be collimated to restrict the field of view to a diameter not greater than 15 cm for adults and proportionately less for children.

To provide a standard sample for determination of thyroid uptake, a vessel resembling the average thyroid (e.g., a 30-ml polyethylene bottle 3 cm in diameter) is filled with an amount of ^{123}I equal to that delivered orally to the patient. This standard sample is placed at a depth of 0.5 cm behind the surface of a neck phantom composed of a cylinder of Lucite 15 cm in diameter and 15 cm tall. The percent uptake of iodine

**FIGURE 12-1**

Pulse height spectra for a 1-ml, 5- μ Ci ^{131}I source in air and in a neck phantom at a distance of 30 cm from a 1.5-in.-diameter, 1-in.-thick NaI(Tl) crystal. Radiation scattered by material around the radioactive source increases the height of the Compton plateau.

in the thyroid is the ratio of the net count rate measured in front of the thyroid to that measured under identical conditions with the detector in front of the neck phantom:

$$\text{Uptake (\%)} = (100) \frac{(c)_t}{(c)_s} \quad (12-1)$$

where c_t is the count rate for the thyroid and c_s is the count rate for the standard.

Radiation scattered from extrathyroidal tissue should be considered in the computation of percent uptake (Figure 12-1). With one method of correction for extrathyroid activity, count rates for the patient and the phantom are measured before and after a lead block (B-block) about 1/2 in. thick has been placed over the thyroid and the radioactive source in the phantom.¹ The percent uptake is

$$\text{Uptake (\%)} = (100) \frac{c_t - c_{tL}}{c_s - c_{sL}} \quad (12-2)$$

where c_t is the count rate for the thyroid, c_{tL} is the count rate for the thyroid shielded by a lead block, c_s is the count rate for a standard, and c_{sL} is the count rate for a standard shielded by a lead block.

Example 12-1

The following count rates were obtained 24 hours after administration of 0.5 MBq of ^{123}I to a patient:

For the patient's thyroid	16,110 cpm
For the patient's thyroid shielded by a lead block	5,790 cpm
For a standard	96,450 cpm
For a standard shielded by a lead block	5,460 cpm

What is the percent uptake?

$$\begin{aligned} \text{Uptake (\%)} &= (100) \frac{c_t - c_{tL}}{c_s - c_{sL}} \\ &= (100) \frac{16,110 - 5,790}{96,450 - 5,460} \\ &= 11\% \end{aligned} \quad (12-2)$$

Sometimes a standard count rate is obtained by counting the activity of the capsule of ^{123}I before it is administered to the patient.

de Hevesy was the “father” of the tracer method of using radioactive isotopes to map biological processes. For this method, he needed relatively long-lived radioactive isotopes of elements that occur naturally in the body. His work received a significant boost in 1934 with arrival of artificially produced ^{32}P ($T_{1/2} = 14.3$ days) from Enrico Fermi's laboratory in Rome.

Dilution Studies

Dilution measurements with radioactive nuclides are used for a variety of diagnostic tests in clinical medicine. If a solution of volume v_i that contains a known concentration $(\text{cpm/ml})_i$ of radioactive material is thoroughly mixed with a large volume of nonradioactive solution, then the volume v_f of the final solution may be estimated by measuring the specific count rate $(\text{cpm/ml})_f$ of the final solution:

$$v_f = \frac{(\text{cpm/ml})_i}{(\text{cpm/ml})_f} v_i \quad (12-3)$$

Because $(\text{cpm/ml})_i v_i$ equals the total counts per minute $([\text{CPM}]_i)$ of the injection solution, Eq. (12-3) may be written as

$$v_f = \frac{(\text{cpm})_i}{(\text{cpm/ml})_f} \quad (12-4)$$

Among the earliest dilution experiments were those of de Hevesy. In 1934, for example, de Hevesy measured his total body water by drinking “heavy water” (water containing deuterium in place of ordinary hydrogen) and measuring the concentration of deuterium in his urine.

The specific count rates $(\text{cpm/ml})_i$ and $(\text{cpm/ml})_f$ must be corrected for background. The difference $v_f - v_i$ is the volume of the nonradioactive solution with which the solution of volume v_i is mixed.

During some applications of the dilution technique, the specific count rate $(\text{cpm/ml})_f$ of the final solution must be corrected for activity that escapes from the volume to be measured. For example, estimates of blood volume with serum albumin labeled with ^{131}I sometimes are made at intervals over the first hour after intravenous injection of the labeled compound. To correct for the escape of serum albumin from the circulation, the specific count rate may be plotted as a function of time and extrapolated back to the moment of injection. Alternately, the blood volume may be measured at a specific time (e.g., 10 minutes after injection of the labeled albumin), and a predetermined correction factor may be applied to the measured volume.

Example 12-2

0.5 MBq of serum albumin tagged with ^{131}I (radioiodinated human serum albumin [RISA]) are injected intravenously into a patient. After the labeled albumin has thoroughly mixed in the patient’s circulatory system, 5 ml of the patient’s blood is withdrawn. The 5-ml sample yields a count rate of 2675 cpm. Another 0.5 MBq sample of RISA is diluted to 2000 ml. A 5-ml aliquot from this standard solution yields a count rate of 6343 cpm. The background count rate is 193 cpm. What is the total blood volume?

$$\begin{aligned} \text{Sample net specific count rate} &= (2675 - 193) \text{ cpm}/5 \text{ ml} \\ &= 496 \text{ cpm/ml} \end{aligned}$$

$$\begin{aligned} \text{Standard net specific count rate} &= (6343 - 193) \text{ cpm}/5 \text{ ml} \\ &= 1230 \text{ cpm/ml} \end{aligned}$$

$$\begin{aligned} \text{Count rate for a 0.5 MBq sample} &= (1230 \text{ cpm/ml}) (2000 \text{ ml}) \\ &= 246 \times 10^4 \text{ cpm} \end{aligned}$$

If the volume of the injected RISA is ignored, the blood volume v_f is

$$\begin{aligned} v_f &= \frac{(\text{cpm})_i}{(\text{cpm/ml})_f} \\ &= \frac{246 \times 10^4 \text{ cpm}}{496 \text{ cpm/ml}} \\ &\approx 5000 \text{ ml} \end{aligned} \quad (12-4)$$

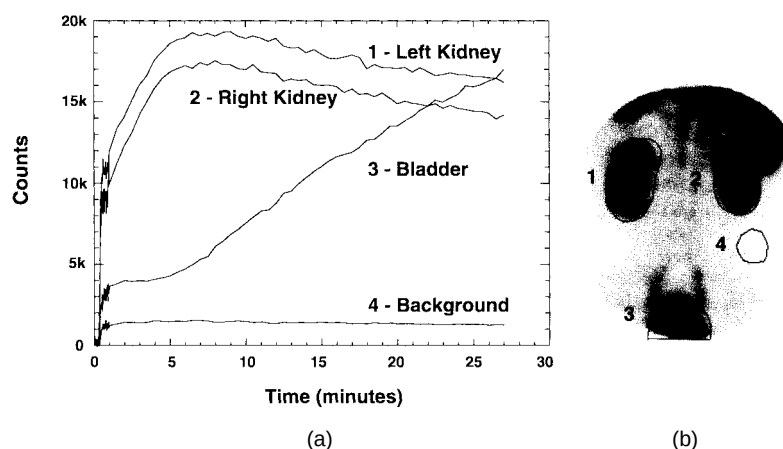


FIGURE 12-2

A: Dynamic renal study showing excretion (changes in count rate as a function of time) of technetium $^{99m}\text{Tc-MAG}_3$. **B:** Regions of interest for left and right kidneys, bladder, and background are drawn from the static image. (Courtesy of Bruce Hasselquist, Ph.D.)

Time–Activity (Dynamic) Studies

Static images display the spatial distribution of radioactivity in an organ in an effort to demonstrate structural abnormalities, reveal physiologic defects, or estimate relative mass. Some nuclear medicine studies have been devised to measure the radioactivity in an organ over time after administration of a radioactive compound. These studies are termed *dynamic studies*. Examples are renal studies to evaluate blood flow and functional renograms to determine kidney function. Dynamic and static studies often involve different radiopharmaceuticals. Static renal studies are performed with ^{99m}Tc -dimercaptosuccinic acid (DMSA) or ^{99m}Tc -glucoheptonate to image the renal parenchyma, whereas effective renal plasma flow is evaluated with ^{131}I -Hippuran or ^{99m}Tc mertiatide ($^{99m}\text{Tc-MAG}_3$). Dynamic studies may include the summation of data over time to provide a “static” image. For example, both static and dynamic renal studies can be acquired with a single injection of ^{99m}Tc -glucoheptonate.

A dynamic renal study obtained with $^{99m}\text{Tc-MAG}_3$ is shown in Figure 12-2. Regions of interest are drawn for the left and right kidneys, the bladder, and background. The activity of ^{99m}Tc is shown as it is excreted by the kidneys via active tubular secretion and glomerular filtration.

■ SINGLE-CRYSTAL SCINTILLATION CAMERA

The first single-crystal scintillation camera was assembled in 1956 by Anger at the University of California.² This development was a major advance in static imaging of radioactivity distributions, because it allowed simultaneous collection of γ rays from all regions of small organs. The camera contained a 6-mm-thick sodium iodide [NaI(Tl)] crystal only 10 cm in diameter. It was viewed by seven photomultiplier (PM) tubes, each 3.8 cm in diameter. Six PM tubes were hexagonally arranged about the seventh tube at the center. In 1960, work began at Nuclear-Chicago Inc. on an improved version of the Anger camera for commercial marketing. In 1963, a larger scintillation camera was built with a 28-cm crystal and 19 PM tubes arranged in hexagonal fashion. In this camera, as in earlier versions, the PM tubes were optically coupled to the scintillation crystal with mineral oil, a technique that furnished efficient transmission of light from the crystal but restricted the position of the detector to an overhead view of the patient. A major improvement in camera design was replacement of the mineral oil by a solid plastic light guide and optical coupling grease. This change eliminated the oil seals required for earlier cameras and permitted rotation of the detector head to any desired position.

The first application of radioactive tracers to experimental medicine were studies by Blumgart, Weiss, and Yens in 1926 that used a decay product of radium to measure the circulation time of blood.

Modern cameras can be used for dynamic as well as static studies.

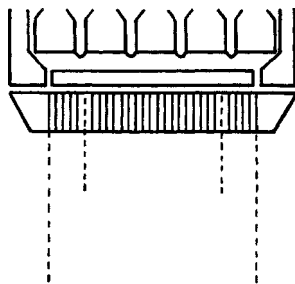
The first commercial scintillation camera was installed at Ohio State University in 1962.

Scintillation cameras are sometimes referred to as gamma cameras or Anger cameras.

The element technetium has no stable isotopes, and consequently does not occur in nature. It was first discovered in 1937 by the Italian physicist E. Segrè in molybdenum heat shields and deflectors discarded from E. O. Lawrence's first cyclotron at Berkeley. Segrè was awarded the 1959 Nobel Prize in Physics for the discovery of the antiproton.

Advantages of ^{99m}Tc for nuclear medicine studies include:

- Short half-life (6 hr)
- Ideal γ -ray energy (140 keV)
- Little particulate radiation
- Ease of labeling
- Administration of relatively high activities



MARGIN FIGURE 12-1
Parallel multihole collimator.

A high-resolution collimator provides relatively low sensitivity, whereas a high-sensitivity collimator yields relatively low spatial resolution.

Although the basic design of the scintillation camera has not changed much since 1963, many improvements have been added that have increased its speed and versatility and enhanced its usefulness in clinical medicine. Among these improvements are replacement of vacuum tubes by integrated circuits and the use of precision electronic components to obtain a more rapid and reliable signal that is less sensitive to environmental and power fluctuations. Currently, scintillation cameras are available with crystals up to 0.5 m in diameter coupled to as many as 91 PM tubes. This expansion has improved the quality of scintillation camera images. Another development that has greatly enhanced the usefulness of the scintillation camera is interfacing of the camera to data storage and processing systems. Other modifications include the use of more efficient bialkali PM tubes, utilization of multichannel analyzers for window selection and adjustment, and incorporation of digital computers into the image acquisition and display network.

The widespread availability of ^{99m}Tc is one factor that has contributed greatly to use of the scintillation camera and to nuclear medicine procedures in general. This radionuclide emits a γ ray of 140 keV, an ideal energy for scintillation camera imaging, with only slight contamination by particles and photons of undesired energies. In addition, ^{99m}Tc has a short half-life (6 hours) and can be incorporated into a variety of compounds and drugs. The short half-life of ^{99m}Tc permits administration of large amounts of radioactive material without excessive radiation doses to patients. Increased administration of radioactivity means that more γ rays are emitted over a shorter period of time, permitting the scintillation camera to be used to its full advantage.

■ PRINCIPLES OF SCINTILLATION CAMERA OPERATION

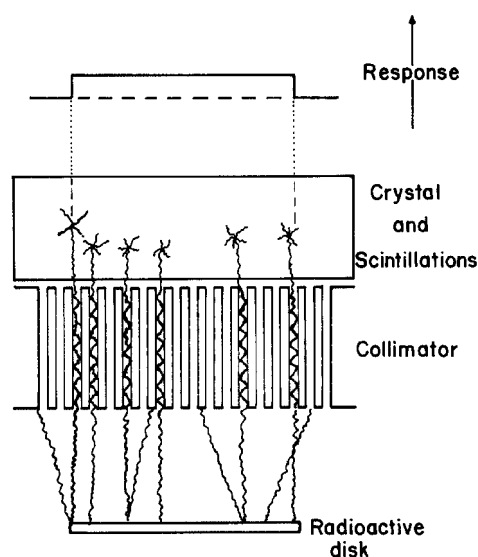
A single-crystal scintillation camera can be separated into four sections: the collimator, the detector, the data processing section, and the display unit.

Collimators

To produce an image of the distribution of radioactivity in a patient, the sites of absorption of γ rays in the scintillation camera crystal must be related clearly to the origin of the γ rays within the patient. This relationship can be achieved by placing a collimator between the crystal and the patient. The most commonly used collimator is the parallel multihole collimator. This collimator is composed of small, straight cylinders separated by lead or tungsten septa (Margin Figure 12-1). Relatively thin septa are used in low-energy collimators designed for use with radionuclides emitting low-energy photons. Thicker septa are required for collimators used with nuclides emitting higher-energy photons. The ability of high-energy photons to traverse even relatively thick septa establishes an upper limit of about 600 keV on the photon energy range useful for scintillation camera imaging. As a general rule, the minimum path length through septa of the collimator should be at least five mean free paths ($<1\%$ transmission) for the photons with which the collimator is to be used.

A γ ray moving from the patient toward the crystal in a direction in line with one of the collimator cylinders can easily pass through the cylinder and interact in the crystal. However, a γ ray emerging obliquely from the patient must traverse one or more of the lead septa in the collimator before reaching the crystal (Figure 12-3). Hence, γ rays emitted obliquely have a much lower probability of reaching the detector. By this mechanism, γ rays interacting in the crystal can be related spatially to their origin in small discrete volumes of tissue directly in front of the collimator.

A collimator with many cylinders of small diameter is termed a *high-resolution collimator*. A collimator with fewer cylinders of larger diameter, or with shorter cylinders of the same diameter, allows more photons to reach the crystal and is termed a *high-sensitivity collimator*. In the selection of parallel multihole collimators, a gain in sensitivity is accompanied by a loss of resolution and vice versa.

**FIGURE 12-3**

With a parallel multihole collimator, γ rays emitted in line with the collimator holes are transmitted to the crystal, whereas γ rays emitted obliquely are absorbed by the collimator septa.

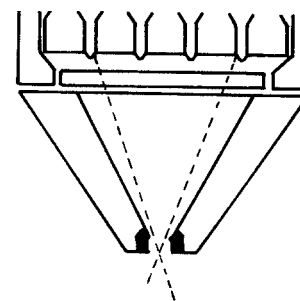
As the diameter of the collimator cylinders is reduced, fewer photons are admitted through each cylinder into the scintillation crystal. Hence, the statistical influence of variations in photon emission (i.e., quantum noise or mottle) is increased. The only way to prevent an increase in image noise is to increase the imaging time or the emission from the source. Either of these changes ensures that the number of photons passing through each cylinder of the collimator remains constant. As an example, suppose that the resolution of a nuclear medicine imaging system is improved by a factor of 2 by decreasing the diameter of the collimator cylinders. Also suppose that this improvement is accompanied by a twofold reduction in the number of photons reaching the crystal. To keep the quantum noise of the image at the original level, the number of photons incident upon the collimator must be increased by a factor of 2. To reduce the quantum noise even further to a level corresponding to the finer resolution of the new collimator, however, an additional fourfold increase would be required in the number of photons incident upon the collimator. That is, a twofold improvement in the spatial resolution of an imaging system achieved by reducing the diameter of the collimator cylinders should be accompanied by an eightfold increase in photon emission or imaging time to obtain a parallel improvement in image noise.

As the distance is increased between the face of a parallel multihole collimator and the patient, the resolution decreases because each open cylinder in the collimator views a larger cross-sectional area of the object. For the same reason, the sensitivity does not change significantly with the distance between a multihole collimator and the patient. Although the number of photons collected by each collimator cylinder from a unit area of the patient decreases with the square of the collimator-object distance, the area viewed by each cylinder increases as the square of the distance. These two influences cancel each other so that the sensitivity does not change appreciably with alterations in the collimator-patient distance.

Other types of collimators used occasionally with scintillation cameras include the pinhole collimator, the diverging collimator, and the converging collimator.

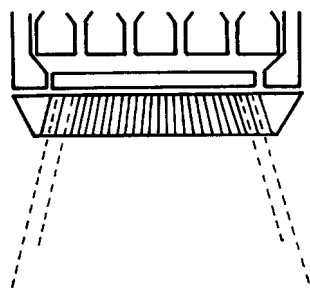
The pinhole collimator consists of a pinhole aperture in a lead, tungsten, or platinum absorber. It is used primarily to obtain high-resolution images of small organs (e.g., the thyroid) that can be positioned near the pinhole. The major disadvantage of the pinhole collimator is its low sensitivity because only a few of the photons emitted by the object are collected to form an image. Occasionally, a pinhole collimator with three apertures is used to provide three different views of a small organ simultaneously.

The spatial resolution of a scintillation camera is termed the *intrinsic resolution*. Its value is typically 4–5 mm. This resolution is degraded significantly in practice by the presence of a collimator. The spatial resolution with a collimator in place is referred to as the *extrinsic resolution*.

**MARGIN FIGURE 12-2**

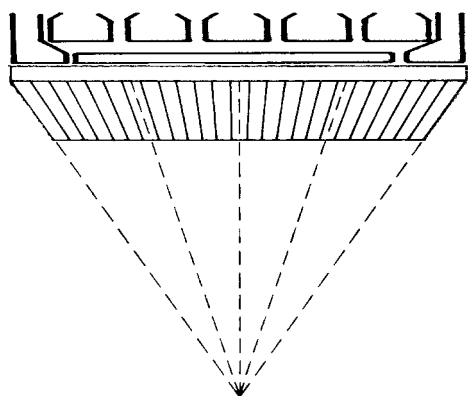
Pinhole collimator.

In the early years of single-photon emission computed tomography (SPECT), a multihole collimator was used for nuclear tomographic imaging. This approach has been superseded by other tomographic approaches.



MARGIN FIGURE 12-3
Diverging collimator.

Collimators that can be used in both converging and diverging modes are referred to as *div/con* collimators.



MARGIN FIGURE 12-4
Converging collimator.

The diverging collimator permits the scintillation camera to image anatomic regions larger than the dimensions of the crystal. The large crystals of up to 20 in. in diameter available with modern scintillation cameras have greatly alleviated the need for diverging collimators. When compared with a parallel multihole collimator, there is some reduction in resolution with a diverging collimator.

With the converging collimator, images with improved resolution can be obtained for organs smaller than the area of the crystal. The resolution is improved because the object is projected over the entire crystal face rather than onto a limited region of the crystal. Also, the photon collection efficiency is greater for a converging collimator than for a parallel multihole collimator used to image a small object. Hence, the influence of quantum noise is often less noticeable in images obtained with a converging collimator.

Coded Aperture Substitutes for Collimators

Conventional collimators provide the necessary positional correspondence between the interaction of a γ ray in the crystal and the origin of the γ ray in the patient. However, they are inefficient collectors of γ rays because they reject all γ rays except those arising within a relatively small solid angle. Also, conventional collimators are often the major limitation in the spatial resolution of a scintillation camera image. As substitutes for conventional collimators, various coded apertures have been proposed. One of the simplest of these devices is the Fresnel zone plate. With this device, each point of radioactivity in the patient casts a unique shadow of the zone plate onto the detector (Figure 12-4). That is, points of radioactivity at the same distance from the zone plate cast shadows on the detector that are laterally displaced from each other, and points at different distances from the zone plate cast shadows of different sizes. Because of these effects, information is projected onto the detector about the three-dimensional distribution of radioactivity in the patient.

To retrieve this information, the transmission data projected onto the detector may be recorded in coded form on film. When this coded image is placed in a coherent light beam, diffraction of the light produces an interpretable image that can be viewed telescopically (Figure 12-5). This image is tomographic, and movement of the eyepiece of the telescope allows one to visualize the distribution of radioactivity at different depths in the patient.

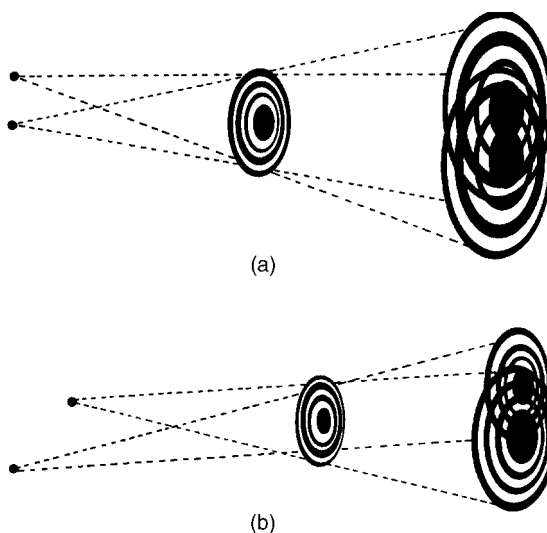


FIGURE 12-4

In zone plate imaging, each point of radioactivity casts a unique shadow onto the detector. (From Farmelant, M., et al.³ Used with permission.)